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An interpretable machine-learning pipeline for deforestation risk prediction in Vietnam, at 1 km resolution



PRESENTED BY “V I G I L” TEAM:
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THE PROBLEM

01 Vietnam's forests face persistent loss where land pressure, access, and governance limits collid

02 Monitoring today is reactive - it records loss only after it happens

03 Existing risk-mapping tools aren't reproducible, aren't tested across regions, and are hard for non-technical rangers to act on

04 Forest-protection resources are thin relative to the area needing patrol

05 Satellite loss data alone can't tell legal logging from illegal clearing



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VIGIL'S VISION

- **Goal:** a complete, nationwide early-warning pipeline - from data to trained models to public alerts for rangers

- **Three defining traits:**
  - + Nationwide by design - not one-province; interpretable
  - + Explainable - every prediction traceable to a feature
  - + Full-lifecycle - data → training → deployment → field feedback



OUR CONTRIBUTIONS AT A GLANCE

A general grid-based pipeline on open geographic + climatic data

Baseline vs. ML, tested two ways: same-area split and cross-district transfer

Capture-rate metrics that turn rankings into patrol priorities

Explainable outputs built for non-technical stakeholders

A live, province-agnostic public platform with crowdsourced verification

A concrete path to nationwide scale

THE 3-PILLAR DATA ARCHITECTURE

- A model general enough for nationwide use needs three data pillars

ready/in use

Geographic

terrain and baseline forest cover

example: elevation and slope from SRTM, baseline tree cover from Hansen GFC

Fixes where the forest is and what terrain surrounds it - the static baseline every prediction is anchored to

ready/in use

Climatic:

rainfall and vegetation/disturbance signals

example: rainfall from CHIRPS, NDVI/NBR vegetation indices from Sentinel-2

Track rainfall and vegetation/ disturbance signals - the dynamic trigger that flags where risk is rising

awaiting national data

Human/Production-Context:

local logging and forest-management records - separates legal logging from illegal clearing in the ground-truth label

example: district logging permits, forest-management plan boundaries

The only pillar that can separate legal timber harvest from illegal clearing in the ground-truth label

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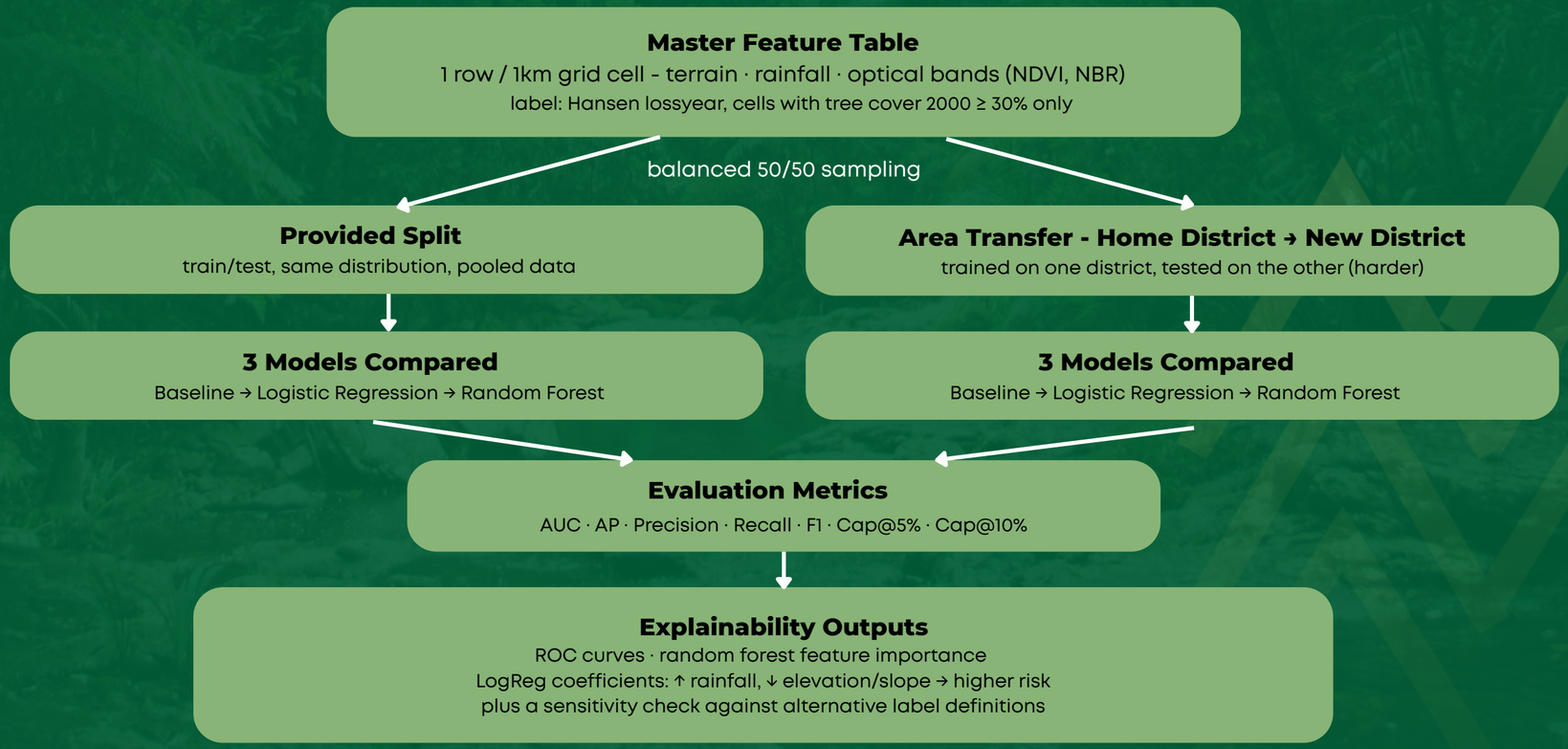
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METHODOLOGY: DATA SOURCES

DATASET	LAYERS	NATIVE RESOLUTION	TIME SPAN	ROLE
4 public datasets, all processed in Google Earth Engine	Hansen Global Forest Change → forest baseline + historical loss	SRTM → terrain (elevation, slope)	CHIRPS → rainfall	Sentinel-2 → vegetation and disturbance indices (NDVI, NBR)

METHODOLOGY:
GRID-BASED MODELING ARCHITECTURE

- A 1 km grid per district; predictors and labels aggregated per cell
- 2 validation tests: a same-area split, and a cross-district transfer test
- Sampling is balanced for stable training
  Also means probabilities need calibration before real deployment



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METHODOLOGY:
PREDICTIVE ALGORITHMS

- **Three models compared:**
  + Baseline (transparent z-score)
  + Logistic Regression (interpretable)
  + Random Forest (captures nonlinear patterns)

- Metrics:
  + AUC
  + precision/recall/F1
  + capture-rate (top 5%/10% highest-risk cells)

➡ **Chosen to span a transparency-to-complexity spectrum; boosting methods (XGBoost, LightGBM) are planned for the national scale-up**

METHODOLOGY: MODELING PIPELINE

- Google Earth Engine handles data access and aggregation; Python trains the models and exports results
- Every step is reproducible and independently verifiable



Data Sources for VIGIL

Four public satellite sources-Hansen GFC, SRTM, CHIRPS, Sentinel-2

Risk Prediction with VIGIL

Baseline, logistic regression, and random forest compared side by side

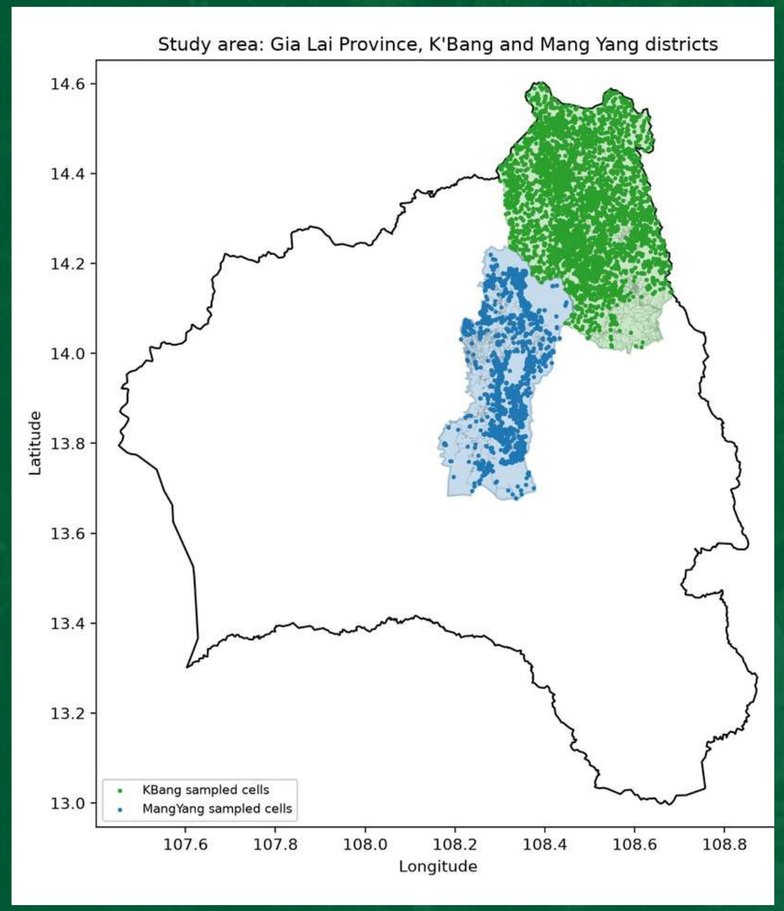
Live Deployment on Public Website

VIGIL is live at www.vigil.green/map, mapping risk zones and collecting field reports today

Operational Integration with Local Rangers

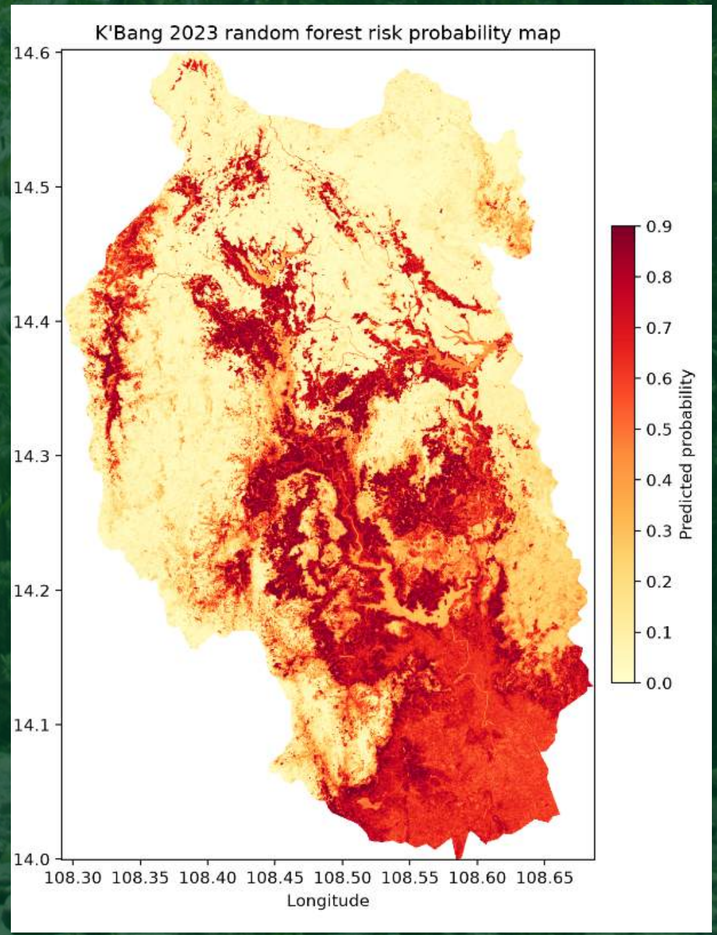
Ranks the riskiest 5% of grid cells before loss occurs and display on live map

Study Area



- Gia Lai province:
  - K'Bang District: training
  - Mang Yang District: transfer test

- The 2 districts were chosen for contrasting forest and terrain conditions



- 6,000 balanced grid cells total: K'Bang 4,000, Mang Yang 2,000

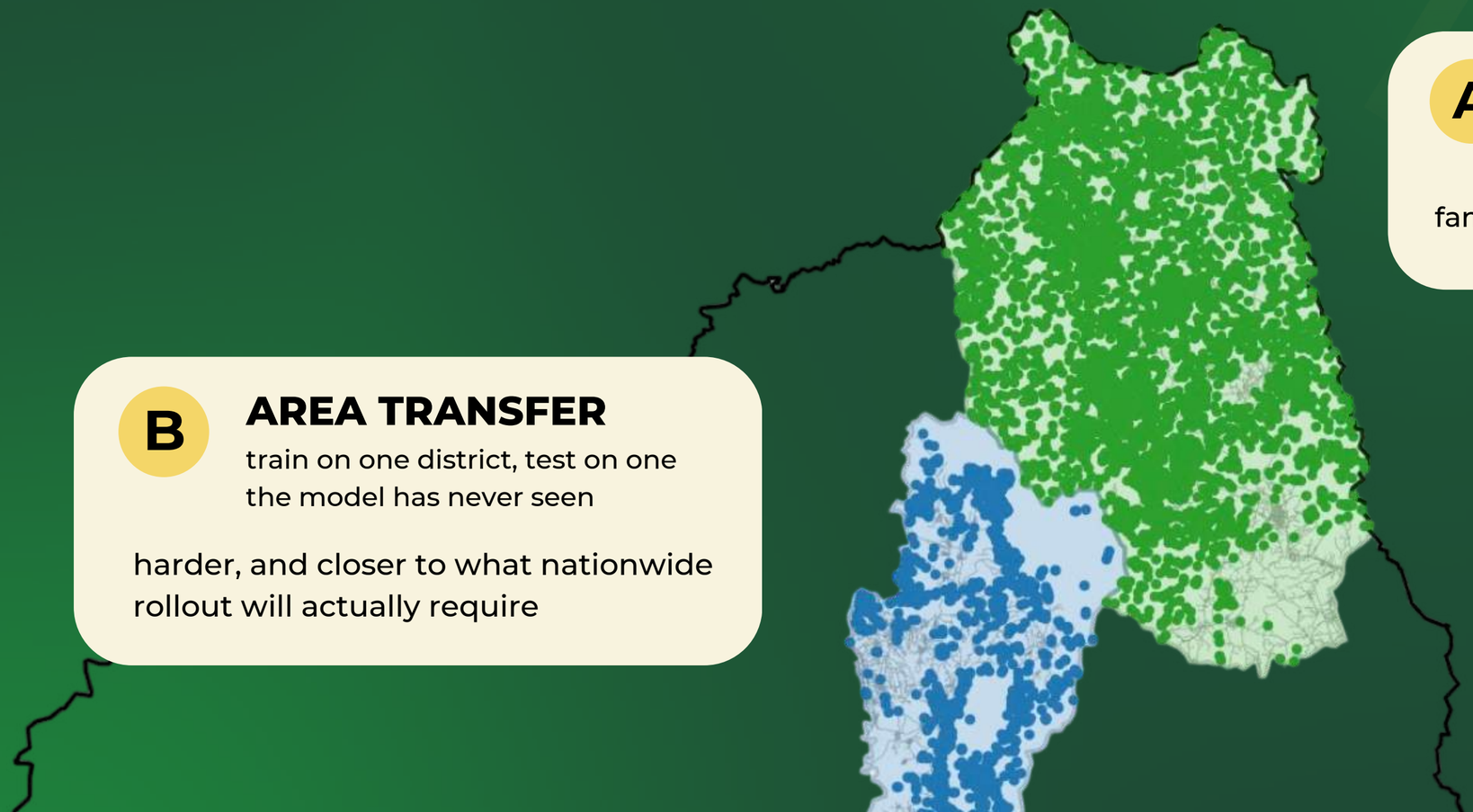
- Split into 4,788 training and 1,212 test cells

Dataset at a Glance

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Two Honest Tests, Not One Convenient Test



A **SAME-AREA SPLIT**

Best-case performance

familiar conditions

B **AREA TRANSFER**

train on one district, test on one the model has never seen

harder, and closer to what nationwide rollout will actually require

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Results: Provided Split

- AUC: Baseline 0.504, LogReg 0.817, RF 0.861
- Cap@5%: 0.077 / 0.089 / 0.087
- Full metrics: AP, precision, recall, F1, Cap@10%, N_test, N_pos

METRIC	BASELINE	LOGREG	RF
AUC	0.504	0.817	**0.861**
F1	0.491	0.771	**0.788**
Cap@5%	0.077	0.089	**0.087**
Cap@10%	0.129	0.179	**0.176**

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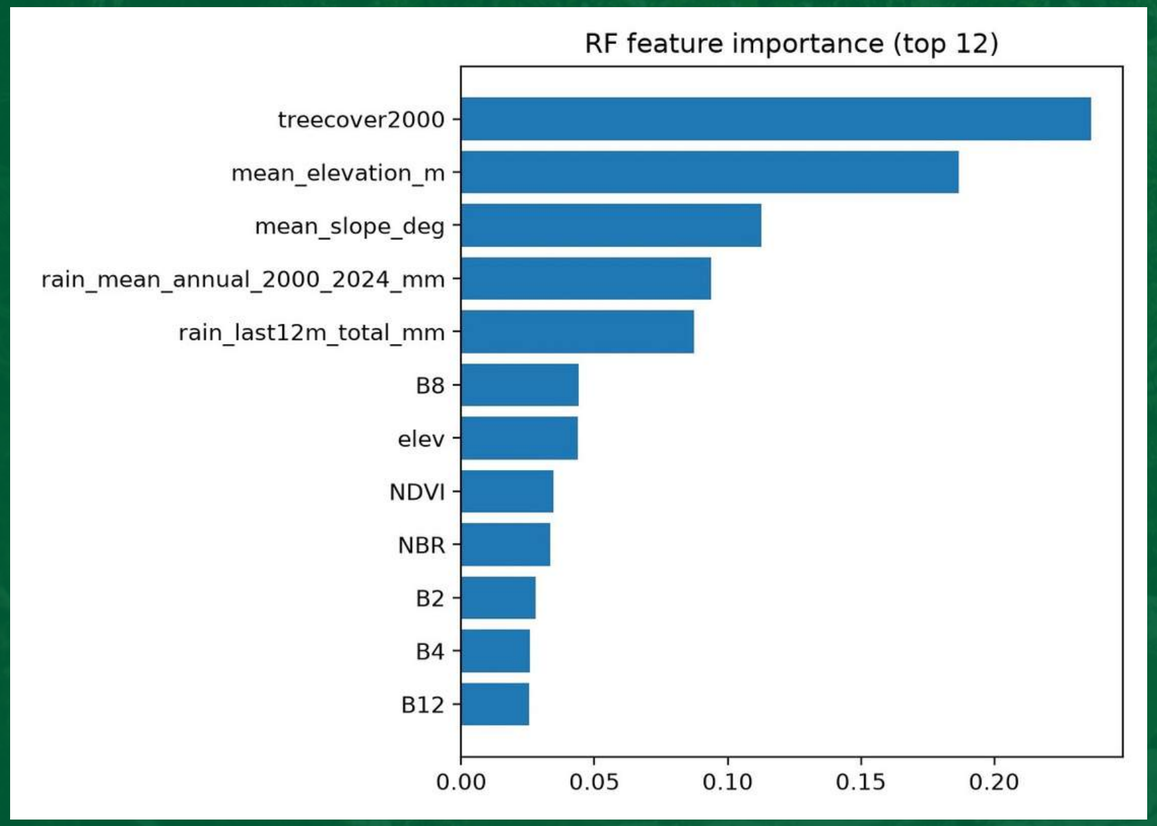
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Results: Area Transfer (K → M)

- AUC: Baseline 0.376, LogReg 0.589, RF 0.535
- Cap@5%: 0.028 / 0.062 / 0.059
- Full metrics: AP, precision, recall, F1, Cap@10%, N_test, N_pos

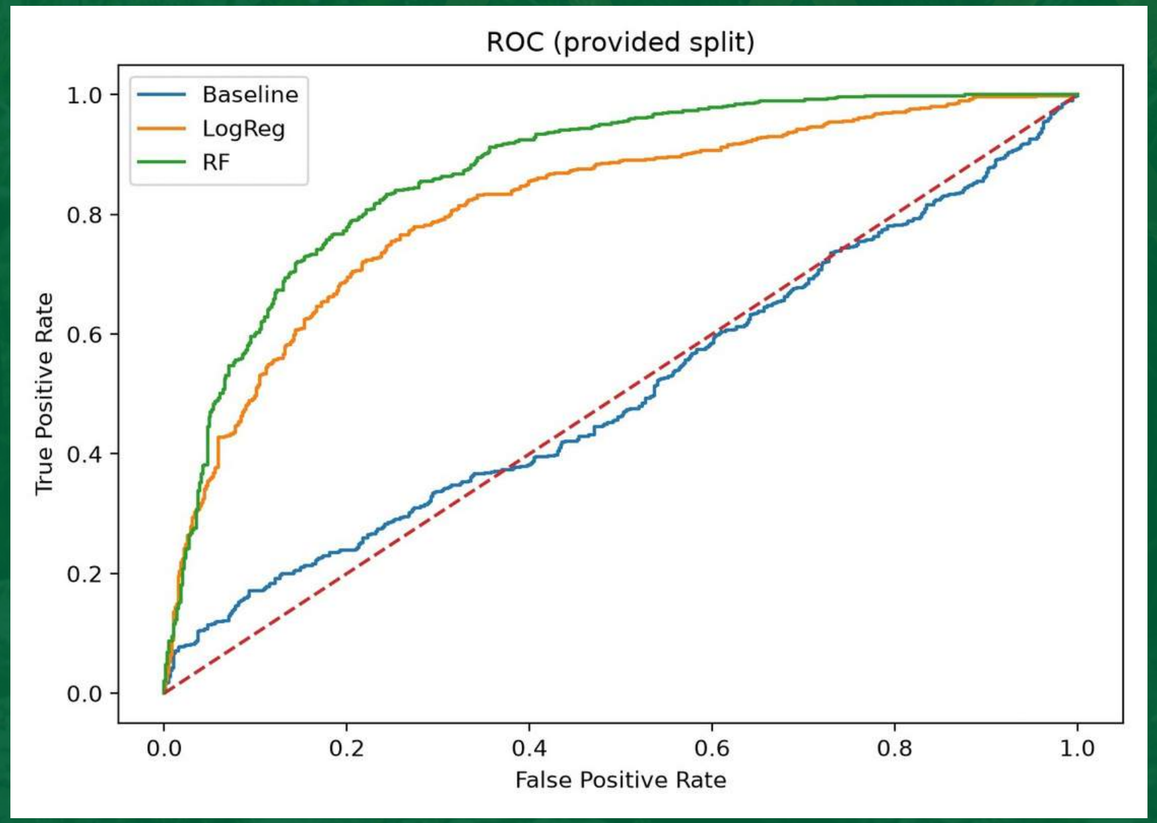
METRIC	BASELINE	LOGREG	RF
AUC	0.376	**0.589**	0.535
F1	0.654	**0.627**	0.593
Cap@5%	0.028	**0.062**	0.059
Cap@10%	0.059	**0.135**	0.111

Curves & Explainability



Random Forest feature

* which inputs (tree cover, elevation/slope, rainfall, spectral bands) drove the predictions most



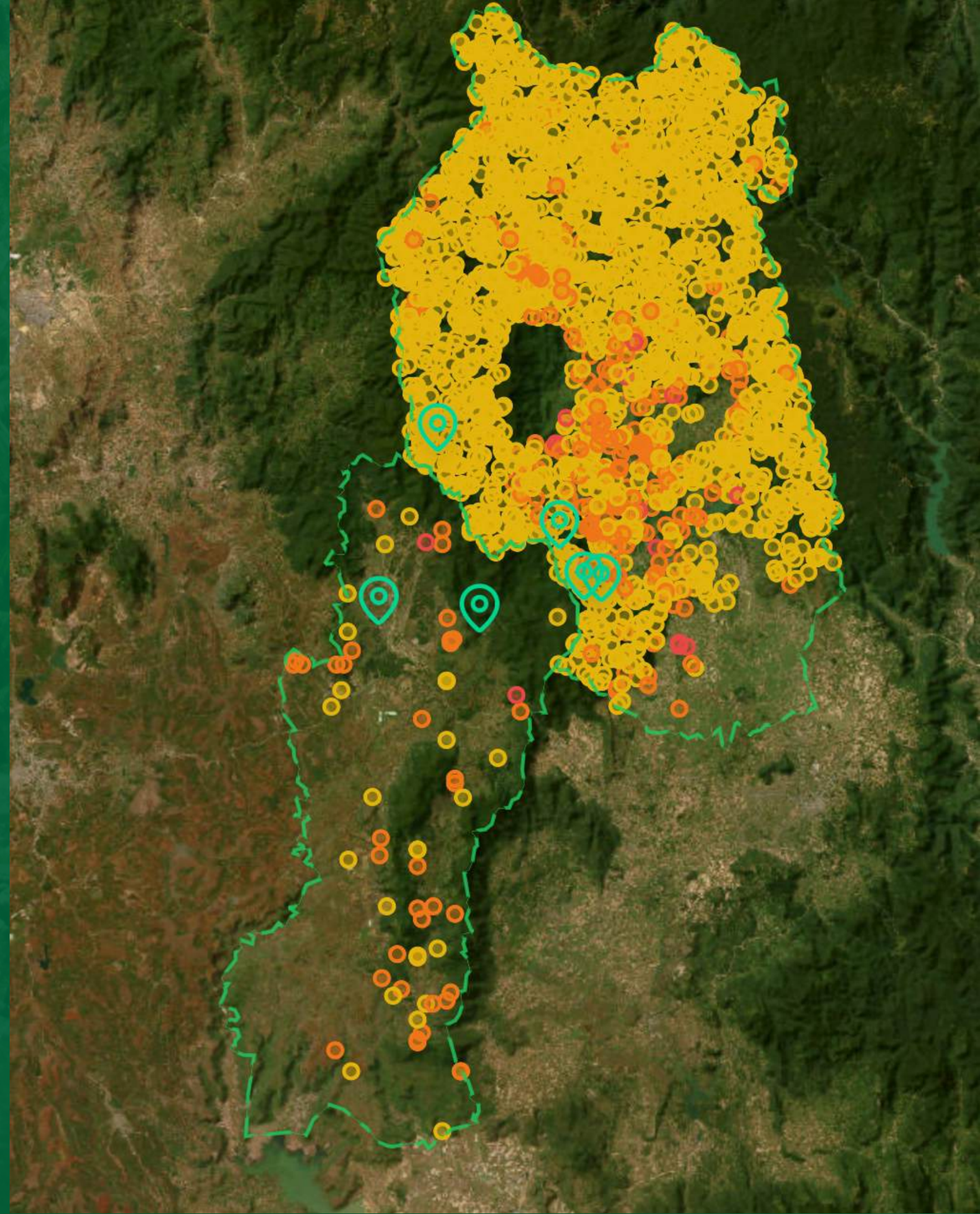
ROC curve

* how well Random Forest, Logistic Regression, and the Baseline separate high- from low-risk cells (Provided Split test)

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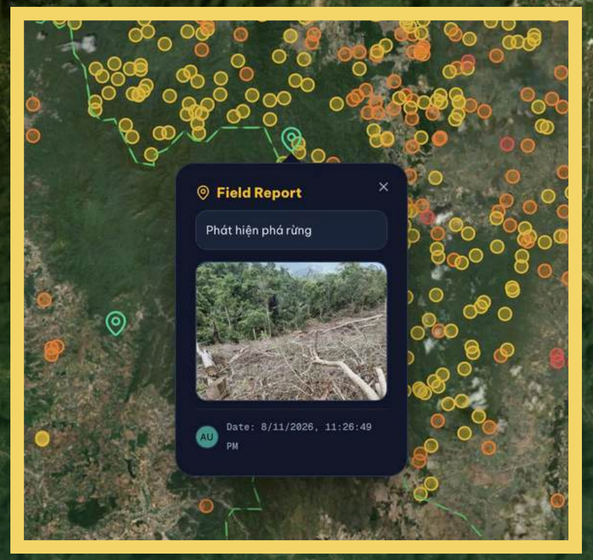
Real-World Impact

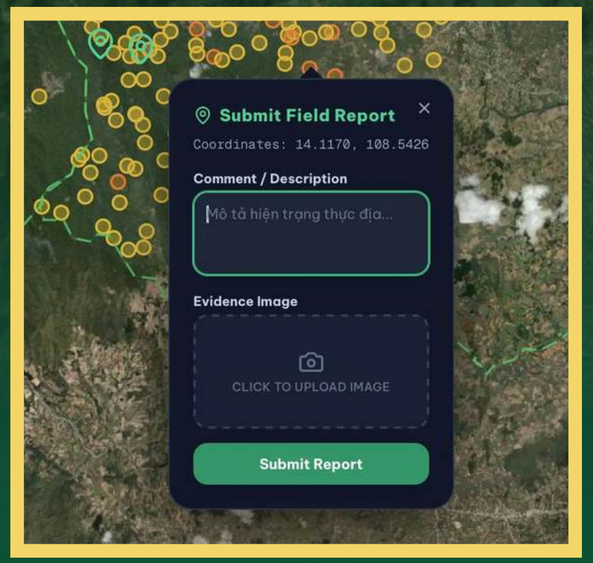
- Top 5%/10% highest-risk cells become ranked patrol targets
- A small share of cells can hold a disproportionate share of actual loss
- Logic: rank all cells → patrol the top-ranked first → re-score as new satellite data arrives
- Complements, rather than duplicates, reactive alert tools like Global Forest Watch



VIGIL Web Platform

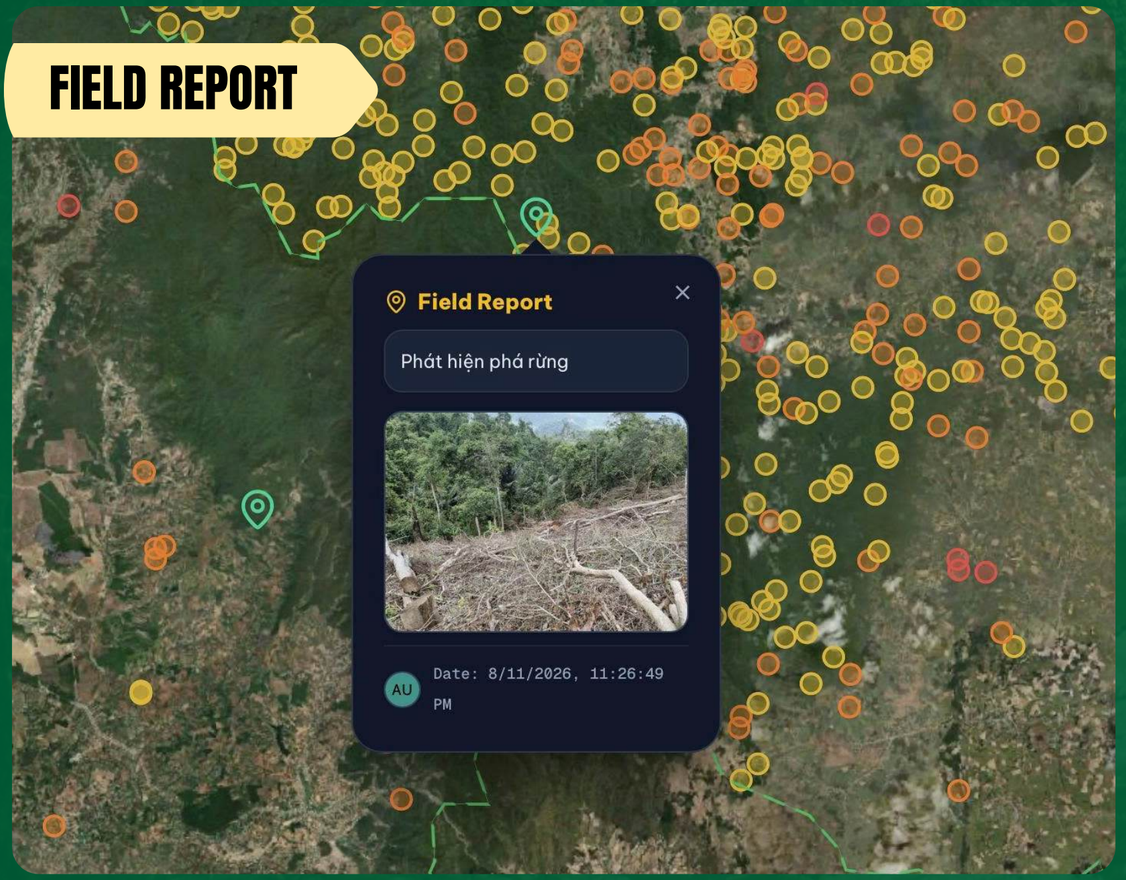
- A public, interactive risk map at vigil.green
- Stack: Next.js/MapLibre frontend, FastAPI + PostgreSQL/PostGIS backend
- Works for any region once modeled - dark/light/satellite views, real-time filters (time, slope, elevation, tree cover, rainfall)
- First live instance covers Gia Lai; displayed metrics are still being finalized against the verified results

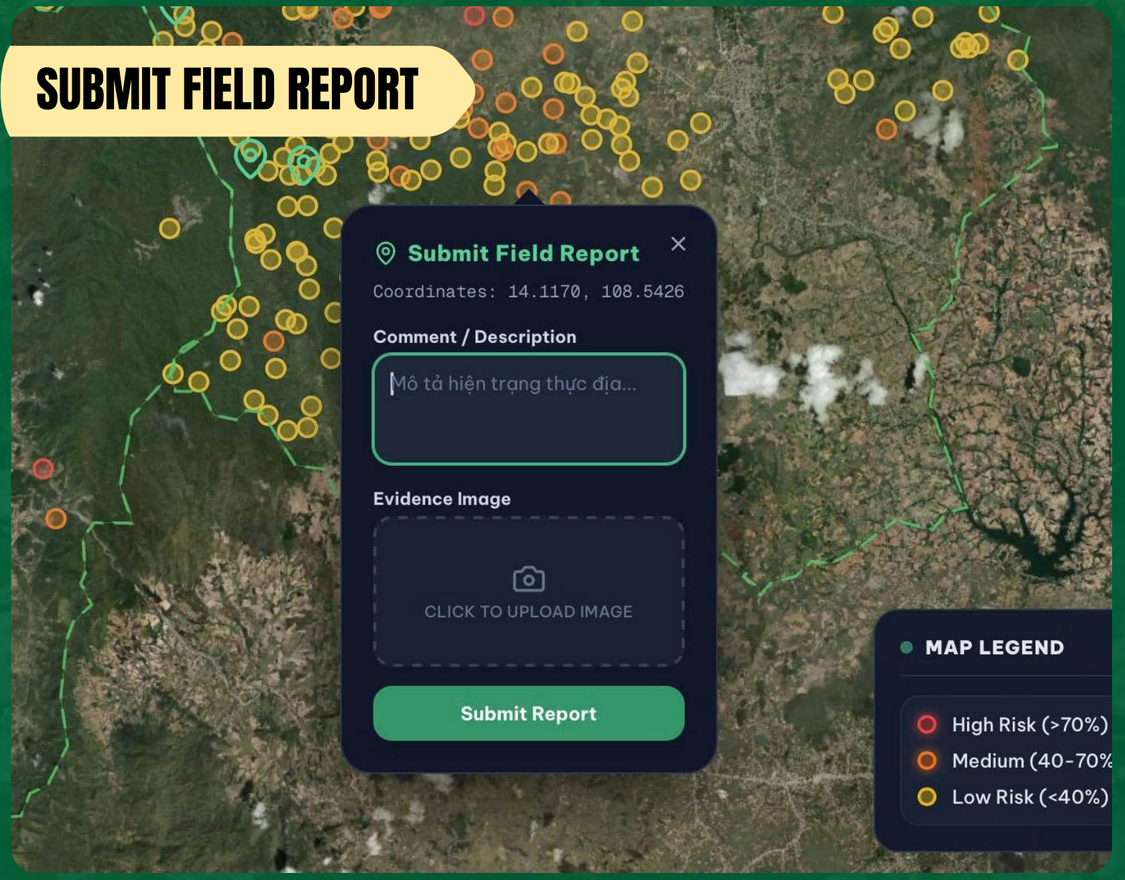




Crowdsourced Field Verification

- Anyone can submit a geotagged field report, with an optional photo
- Reports are moderated before appearing as map markers
- Reporter privacy is protected - hashed IPs, rate-limited submissions
- Not yet feeding back into model retraining - the next milestone





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Limitations

- **Add time-based** validation of when loss actually occurred
- **Add road-access** and land-tenure predictors
- **Surface explainability directly on the live map**
- **Close the loop - feed verified field reports back into retraining**
- **Recalibrate risk probabilities before full-grid operational use**

Ethical Considerations

- Scores land parcels, not people - a resource-allocation tool, not surveillance
- Output is a prioritization aid, never proof of wrongdoing at any location
- Extra caution required outside Gia Lai, where local validation hasn't happened yet
- Reporter privacy protected by design - hashed IPs, rate limits, moderation
- Never used to justify punitive land seizure, displacement, or automated enforcement

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WHY GIA LAI ISN'T THE CEILING ?



- VIGIL's objective was always nationwide coverage
- Gia Lai came first simply because local validation was already possible there

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FUTURE WORK: FOUR-LAYER SCALING MODEL

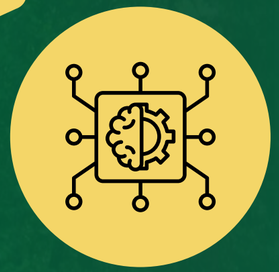
01



DATA

Every new province just needs new grid - inputs are already **public** and **nationwide**

02



MODELING

Repeat the model comparison per region, extended with **boosting methods**, picking the best local performer

03



VALIDATION

Each new province needs its own **local cross-check**, as done for Gia Lai

04



DEPLOYMENT

The platform is already **province-agnostic** - scaling is a data-loading exercise

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VIGIL Business Plan and Investment Case

- Commercialization of locally validated decision-support service for provincial department of forest control with potential of national scale and global expansion
- Optimization of time and resources for the respective clients by pointing out the focused risk area that require more attention

1. Potential Buyers

POTENTIAL CLIENT GROUP	EXAMPLES IN VIETNAM AND WORLDWIDE
Provincial governments and forestry authorities	Vietnam: Provincial Departments of Agriculture and Environment like Gia Lai, Dak Lak, Lam Dong,... Worldwide: equivalent provincial or state forestry authorities.
National forestry and environmental agencies	Vietnam: the Ministry of Agriculture and Environment and its relevant forestry units. Worldwide: national forest services and environmental ministries.
National parks and protected-area management organizations	Vietnam: national-park and nature-reserve management boards. Worldwide: public or independently managed protected areas.
Conservation NGOs	Vietnam: WWF Viet Nam Worldwide: WWF country programmes
Development agencies and programme implementers	Vietnam: GIZ Viet Nam and comparable development programmes. Worldwide: organizations implementing government capacity-building and conservation project.

2. Competition and alternatives

ALTERNATIVE	IMPLICATION FOR VIGIL
Open monitoring platforms and field tools	Prove local prioritization, integration and support value beyond free information
Agency GIS team or local integrator	Compare full operating cost and time to deployment; support standard exports
A bespoke one-off mapping contract	Differentiate recurring data quality, model monitoring and workflow ownership
Existing patrol practice	Run a fair baseline comparison before claiming improved outcomes

3. Commercial offers and boundaries

OFFER	PROPOSED FEE	SCOPE AND BUYING PURPOSE
Data readiness assessment	USD 5,800 one-time	Four to six weeks, data inventory, sample ingestion, feasibility and scoped implementation plan
First provincial pilot	USD 12,000 one-time	Nine-month programme with overlapping development and pilot work in two or three operating areas
New province onboarding	USD 7,700 one-time	Local data setup, bounded configuration, validation setup and user training after the core platform exists
Annual subscription service	USD 15,000 per province/year	Weekly updates, 24-hours support and capped validation effort
Custom or dedicated deployment	Quote after discovery	New connectors, additional areas, high-availability hosting, higher update frequency or special integration

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OUR TEAM MEMBERS

01



PHAM DUY LONG (C)

The Olympia School

02



NGUYEN DUC ANH

Foreign Language
Specialized School

03



DO NHAT QUANG

Vinschool
The Harmony

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Conclusion & Call to Action

01

Proven end-to-end on one real, rigorous pilot

02

Designed from day one to scale nationwide, not "try then decide"

03

Clear next step: FRMS delivery unlocks the four-layer roadmap

04

Contact: vigil.green · Pham Duy Long · Nguyen Duc Anh · Do Nhat Quang

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